

Shuo Sun

Postdoctoral Associate ·
Singapore-MIT Alliance for
Research and Technology
(SMART), M3S
Interdisciplinary Research
Group

He received both his Ph.D. and
B.Eng. degrees in Mechanical
Engineering from the National
University of Singapore, under
the supervision of Professor
Marcelo H. Ang Jr. His work
bridges machine intelligence and
human capability through
robotics, autonomy, and
collective intelligence, with a
particular interest in systems
that make human-AI
collaboration more capable,
meaningful, and deployable.

Email

shuo.sun@smart.mit.edu

LinkedIn

linkedin.com/in/shuosunss

GitHub

github.com/SS47816

Google Scholar

[https://scholar.google.com/citations?](https://scholar.google.com/citations?hl=en&user=xe6YcxYAAAAJ&view_op=list_works&sortby=pubdate)

[hl=en&user=xe6YcxYAAAAJ&view_op=list_works&sortby=pubdate](https://scholar.google.com/citations?hl=en&user=xe6YcxYAAAAJ&view_op=list_works&sortby=pubdate)

ORCID

0000-0002-8432-0452

APPOINTMENTS

2025 - Present

Postdoctoral Associate

Singapore-MIT Alliance for Research and Technology (SMART) · M3S
Interdisciplinary Research Group · Singapore

Research at the intersection of robotics, autonomous systems, and human-AI
collective intelligence.

EDUCATION

2021 - 2025

Ph.D. in Mechanical Engineering

National University of Singapore

Research on autonomous systems, motion planning, prediction, and
simulation under Professor Marcelo H. Ang Jr.

2016 - 2020

B.Eng. in Mechanical Engineering

National University of Singapore

Foundation in robotics, engineering design, and interdisciplinary systems work.

RESEARCH AREAS

- ROBOTICS
- AUTONOMOUS SYSTEMS
- HUMAN-AI COLLECTIVE INTELLIGENCE
- MOTION PLANNING AND PREDICTION
- GENERATIVE SIMULATION

SELECTED PROJECTS

- 2025 - Present

Human-AI Work Intelligence

SMART M3S · Current Research Direction

Current work on intelligent systems that augment human capability, support collective intelligence, and help shape more meaningful AI-enabled labor.
- 2023 - Present

Autonomous Mini-bus

NUS · Tron-E · Turing Drive Taiwan · Flagship Deployment

Development of an autonomous mini-bus platform for on-demand shuttle services in geofenced environments.
- 2020 - 2023

Autonomous Road Sweeper

NUS · Kärcher Germany · Flagship Deployment

A full autonomous cleaning platform for mixed indoor and outdoor environments, from drive-by-wire retrofitting to full-stack software.
- 2023 - 2024

DriveSceneGen

NUS Advanced Robotics Centre · Research System

A generative simulation system that produces diverse and realistic driving scenarios from scratch for autonomous vehicle development and evaluation.

PUBLICATIONS

- preprint

arXiv · 2026

Where can AI be used? Insights from a deep ontology of work activities

Alice Cai and Iman YeckehZaare and Shuo Sun and Vasiliki Charisi and Xinru Wang and Aiman Imran and Robert Laubacher and Alok Prakash and Thomas Malone
- conference

ICRA 2025 · 2025

RMP-YOLO: A Robust Motion Predictor for Partially Observable Scenarios even if You Only Look Once

Jiawei Sun and Jiahui Li and Tingchen Liu and Chengran Yuan and Shuo Sun and Zefan Huang and Anthony Wong and Keng Peng Tee and Marcelo H. Ang

journal

IEEE Robotics and Automation Letters · 2024

DriveSceneGen: Generating Diverse and Realistic Driving Scenarios From Scratch

Shuo Sun and Zekai Gu and Tianchen Sun and Jiawei Sun and Chengran Yuan and Yuhang Han and Dongen Li and Marcelo H. Ang Jr.

conference

IROS 2023 · 2023

FISS+: Efficient and Focused Trajectory Generation and Refinement using Fast Iterative Search and Sampling Strategy

Shuo Sun and Jie Chen and Jiawei Sun and Chengran Yuan and Yuanchen Li and Tangyike Zhang and Marcelo H. Ang

conference

ITSC 2023 · 2023

Maximum Entropy Inverse Reinforcement Learning Based on Frenet Frame Sampling for Human-like Autonomous Driving

Tangyike Zhang and Shuo Sun and Jiamin Shi and Shitao Chen and Marcelo H. Ang and Jingmin Xin and Nanning Zheng

conference

ICARCV 2022 · 2022

Autonomous Research Platform for Cleaning Operations in Mixed Indoor & Outdoor Environments

Shuo Sun and Tangyike Zhang and Zhihai Xiang and Yuhang Han and Dongen Li and Jianghao Li and Zhiyang Liu and Marcelo H. Ang

journal

IEEE Robotics and Automation Letters · 2022

FISS: A Trajectory Planning Framework Using Fast Iterative Search and Sampling Strategy for Autonomous Driving

Shuo Sun and Zhiyang Liu and Huan Yin and Marcelo H. Ang

MENTORING

2019 - Present

Student supervision and mentoring

Mentored more than 70 student theses, capstone, and research projects across robotics and autonomous systems.